
AXIOM: A FALSIFICATION-FIRST REASONING SCAFFOLD FOR CROSS-DOMAIN SYSTEMS ANALYSIS

A PREPRINT

William Battrick
Independent Researcher
Australia
williambattrick@gmail.com

April 12, 2026

ABSTRACT

Large language models produce fluent reasoning that is difficult to audit and vulnerable to proxy optimisation failures: cases where a measurable metric is won while the underlying valued outcome degrades (Goodhart’s Law). Existing structured prompting methods optimise for task performance but do not enforce *measurement integrity* – the explicit separation of what is measured from what is valued – as a required structural property of the output.

We present AXIOM, a falsification-first reasoning scaffold that enforces this separation at the system definition stage, requires at least one explicit falsifier per hypothesis, specifies control loops in four components (signal, comparator, actuator, delay), and audits for metric substitution risk via a mandatory Goodhart check at each analytical step.

We evaluate AXIOM in a blinded generation experiment across 15 operationally distinct domains, comparing structured AXIOM outputs against unstructured baselines. Post-hoc scoring against peer-reviewed literature used a differential-weight rubric emphasising Mechanism Recall ($\times 3$) and Measurement Integrity and Goodhart Handling ($\times 3$), with additional dimensions for Falsifier Coherence ($\times 2$), Experiment-Form Convergence ($\times 2$), and Prediction Consistency ($\times 1$).

AXIOM achieved a weighted score of 93% (461/495), with highest performance on Measurement Integrity (98%) and Falsifier Coherence (96%), and weakest performance on Prediction Consistency (73%) – a dimension not central to the framework’s design purpose under blinded generation conditions. A cross-domain structural invariant was confirmed: the same administrative closure failure pattern appeared independently across six of the 15 domains, confirmed by domain-specific peer-reviewed literature in each case. One genuine out-of-sample knowledge gap was identified (cold-chain freeze damage not detected by vaccine vial monitors), providing a concrete specification for directed extension of the perturbation test procedure.

We argue that measurement integrity, falsification, and Goodhart resistance can be enforced as structural output requirements without domain-specific tuning, and that AXIOM is appropriately positioned as a pre-analysis layer in LLM reasoning architectures. Significant methodological limitations – self-evaluation bias, single-sample generation, and post-hoc weight revision – mean the quantitative results should be treated as exploratory, and the paper establishes the experimental protocol for a confirmatory follow-on study.

Keywords systems analysis · Goodhart’s Law · falsification · reasoning scaffold · LLM governance · measurement integrity · structured reasoning · proxy optimisation · cross-domain validation · LLM reasoning architecture

1 Introduction

Large language models can generate fluent, coherent-sounding analysis across virtually any domain. The problem is not fluency, it is auditability. When a model produces an analysis, it is rarely possible to determine whether it has

correctly identified the gap between what is *measured* and what is *valued*, whether its causal claims are falsifiable, or whether its recommended intervention could be gamed in ways that win the stated metric while degrading the underlying outcome. These are not edge cases. They are the structural failure modes that the literature on Goodhart’s Law, systems theory, and AI specification gaming has documented across decades and domains.

This paper presents AXIOM, a falsification-first reasoning scaffold designed to enforce measurement integrity as a structural output property rather than leaving it as an emergent feature of scale or prompting style. We describe the framework, report a blinded generation experiment validating its outputs against peer-reviewed literature across 15 operationally distinct domains, and discuss the implications for LLM reasoning architecture and organisational systems analysis. (Section 2.9 declares the methodological limitations of the experiment explicitly and in full.)

1.1 Goodhart’s Law and Its Formalisations

The observation that measures become unreliable once they are used as targets was first articulated by Goodhart in the context of monetary policy Goodhart [1975]: once a statistical regularity is exploited for control, it ceases to hold. Strathern’s Strathern [1997] restatement – “when a measure becomes a target, it ceases to be a good measure” – broadened the principle to social and organisational settings and provided the canonical formulation cited in contemporary literature.

Systematic formalisation has developed along two lines. Manheim and Garrabrant Manheim and Garrabrant [2019] classify Goodhart failures into four types: regressional (imperfect proxies), extremal (distributional shift under strong optimisation), causal (severing causal relationships), and adversarial (deliberate gaming). Krakovna et al. Krakovna et al. [2020] survey specification gaming in reinforcement learning as an empirical instantiation of Goodhart’s Law in AI systems. More recently, formal game-theoretic treatments have characterised reward hacking as an equilibrium outcome under finite evaluation capacity: if task quality is multi-dimensional and the evaluation signal is a projection onto a lower-dimensional subspace, distortion is mathematically inevitable regardless of the form of the evaluation function Schmid et al. [2026].

Empirical work in organisational settings confirms that Goodhart effects are not aberrant but structural. Levy et al. Levy et al. [2023] show that in organisational analytics, communication processes make Goodhart failures *inevitable* rather than merely possible, because the representation of computation mediates the performativity of individual action at every stage from data collection to decision-making.

Campbell’s Law Campbell [1979], developed independently in the context of social programme evaluation, provides a parallel formulation with strong empirical support in educational settings: the more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it is to distort the social processes it was intended to monitor.

Despite this substantial literature, two gaps remain. First, formalised Goodhart accounts typically treat the measurement/value gap as a single-layer phenomenon. AXIOM identifies a second structural variant – *nested* Goodhart failure via compounding proxy chains – in which the defence of an intervention refers to a second system whose own measurement architecture is itself Goodhart-compromised (Section 2.4). Second, existing frameworks do not provide a domain-general procedural protocol for detecting these failures in advance of optimisation pressure. AXIOM’s Goodhart check and proxy chain audit are designed to fill this gap.

1.2 Systems Thinking Frameworks

The general theory of systems as a domain-general analytical lens originates with Ludwig von Bertalanffy von Bertalanffy [1968], who formalised open/closed system distinctions, equifinality, and hierarchical organisation. The cybernetics tradition, developed by Wiener Wiener [1948], introduced the feedback loop as the primary analytical unit and established the control loop specification – signal, comparator, actuator, delay – that AXIOM incorporates directly into its core procedure.

Second-order cybernetics (von Foerster von Foerster [1981]; Beer Beer [1972]) added the principle that the observer is structurally part of the system, and Beer’s Viable System Model provided a recursive organisational architecture for autonomous adaptive systems. Meadows Meadows [2008] extended systems thinking for practitioner audiences, introducing stock-and-flow modelling, leverage points, and system archetypes, recurring failure patterns that recur independently across domains. The archetype concept is a direct methodological predecessor to AXIOM’s named structural patterns.

Checkland’s Soft Systems Methodology (SSM) Checkland [1999] addressed a structural limitation of hard systems approaches: in ill-defined, socially constructed problem situations, the boundary of the system of interest is not ob-

jective but perspective-dependent. SSM’s CATWOE analysis and rich picture techniques provide tools for surfacing stakeholder-dependent boundary framings. AXIOM differs from SSM in treating boundary definition as a *prerequisite* for the analytical loop rather than as an iterative negotiation and in requiring falsifiable hypotheses rather than stakeholder consensus.

Senge Senge [1990] operationalised systems archetypes for management settings and identified “shifting the burden” and related failure patterns in organisational systems. Causal loop diagrams (CLDs) Sterman [2000] provide a graphical formalism for these feedback structures. Recent empirical work confirms that CLDs improve systemic reasoning and identification of leverage points in complex problem-solving tasks Podlesnik et al. [2025].

AXIOM’s relationship to this tradition is one of selective integration: it incorporates the control loop specification from cybernetics, the leverage point and archetype concept from Meadows and Senge, and the boundary-condition awareness of SSM. It departs from this tradition in two respects: it places falsification as the central analytic goal rather than systemic understanding, and it is designed explicitly for LLM-assisted reasoning rather than human facilitation.

1.3 Structured Reasoning in Large Language Models

The literature on improving LLM reasoning via structured prompting has grown rapidly since the demonstration that chain-of-thought (CoT) prompting; eliciting intermediate reasoning steps before a final answer; substantially improves performance on multi-step reasoning tasks Wei et al. [2022]. Subsequent work established that zero-shot CoT (“think step by step”) generalises across tasks Kojima et al. [2022], and that self-consistency over multiple sampled reasoning paths further improves reliability Wang et al. [2023].

ReAct Yao et al. [2023a] introduced an interleaving of reasoning traces and tool-use actions, demonstrating that grounding reasoning in external observations reduces hallucination and increases task completion. Tree-of-Thought Yao et al. [2023b] extended this to branching search over reasoning paths with deliberate backtracking. Reflexion Shinn et al. [2023] added verbal reinforcement: agents critique their own prior outputs and revise their reasoning accordingly.

Structured output contracts; requiring LLMs to produce typed, schema-validated JSON rather than free text; have emerged as a complementary approach for production systems where auditability and downstream processing are required OpenAI [2024]. Constitutional AI Bai et al. [2022] introduced a prompting methodology in which models critique and revise their own outputs against a set of stated principles, providing a precedent for principle-driven reasoning architecture.

Despite this progress, several limitations remain. First, existing structured prompting methods optimise for task performance (accuracy, completion rate) but do not enforce *measurement integrity*: they do not require the model to distinguish between what is measured and what is valued, or to identify conditions under which the current response would be falsified. Second, evaluation typically occurs within a single domain or task type; cross-domain transfer of structured reasoning has received limited systematic attention. Third, with the exception of Constitutional AI’s principle layer, existing methods lack a Goodhart check: they provide no structural mechanism to detect when a model is winning the evaluation metric while degrading the underlying epistemic quality of the reasoning.

AXIOM addresses these gaps by treating measurement integrity, falsification, and Goodhart resistance as *required structural properties* of a valid reasoning output, rather than emergent properties of scale or task-specific tuning.

1.4 Cross-Domain Validation of Reasoning Frameworks

Validating reasoning frameworks across domains poses distinctive methodological challenges that differ from standard benchmark evaluation. Standard benchmarks (e.g. MMLU Hendrycks et al. [2021], TruthfulQA Lin et al. [2022], HellaSwag Zellers et al. [2019]) measure performance on fixed test sets, which are vulnerable to training data contamination and to the saturation problem: as model performance approaches ceiling, benchmark scores lose discriminative validity Miranda et al. [2025]. The Putnam-AXIOM benchmark Miranda et al. [2025] demonstrates that models achieving high accuracy on static sets show substantial accuracy drops on programmatically generated variants, confirming that benchmark performance can decouple from underlying capability. Putnam-AXIOM Miranda et al. [2025] is unrelated to the AXIOM framework described in this paper; the name overlap is coincidental.

Cross-domain transfer in reasoning frameworks has a longer history in cognitive science and science education. Far transfer; the application of a learned principle to a structurally similar but superficially different domain; is theoretically possible but empirically rare under typical instructional conditions Barnett and Ceci [2002]. The structural similarity hypothesis holds that transfer depends on recognising deep relational structure rather than surface features Gentner and Holyoak [1997], which is consistent with AXIOM’s explicit abstraction step.

In the LLM literature, cross-domain evaluation remains uncommon. Most evaluations test a single model on a single benchmark or a set of benchmarks within a related cluster (mathematical reasoning, commonsense reasoning, scientific QA). Systematic blinded generation across operationally distinct domains - followed by post-hoc scoring against domain-specific peer-reviewed literature - has, to the authors’ knowledge, not been previously reported as a validation methodology for a reasoning framework.

The self-evaluation bias problem - scoring a framework’s outputs using the same system that generated them - is a known limitation of LLM evaluation Zheng et al. [2023] and is explicitly acknowledged in our Methodological Constraints on Record (Section 2.9). LLM-as-judge methodologies Zheng et al. [2023] have been shown to correlate with human evaluation at the aggregate level while exhibiting systematic biases (verbosity preference, position bias, self-preference). The AXIOM benchmark BackOnTruck et al. [2024] proposes rule-based perturbation and multi-source quality calibration to address these biases; we adopt the transparency approach of declaring the constraint explicitly rather than claiming to have resolved it.

2 The AXIOM Framework

AXIOM (a provisional acronym for *Analytical eXamination via Invariants, Objectives, and Measurement*) is a structured reasoning scaffold designed to improve clarity, reduce wasted effort, and increase transfer of learning across domains by explicitly surfacing the distinction between what is measured and what is valued, the constraints that bound a system, and the conditions under which a current model would be falsified.

AXIOM is not a doctrine or belief system. It is a structured thinking scaffold. The framework is intentionally lightweight and portable: it is designed to reduce cognitive friction, not add to it.

2.1 Design Principles

Six design principles govern AXIOM’s construction and intended application.

1. **Provisionality.** AXIOM is a testing scaffold, not a worldview. All models produced by its application are explicitly temporary and subject to revision when falsifying evidence is produced.
2. **Structural primacy.** AXIOM separates narrative framing, structural incentives, resource constraints, and coupling relationships before drawing conclusions. Structure is analysed before mechanism; mechanism before interpretation.
3. **Falsification over confirmation.** Disconfirming tests are preferred to confirming ones. The framework requires that every hypothesis be accompanied by at least one explicit falsifier: a statement of what observable outcome would demonstrate that the hypothesis is wrong.
4. **Cross-domain transfer.** Once a structural pattern is identified in one domain, AXIOM abstracts it and tests its applicability in others. Transfer of pattern is treated as evidence of structural generality.
5. **Biological constraint layer.** AXIOM is explicitly suspended when fatigue, dysregulation, high cognitive load, or emotional activation is present. Physiology is treated as a hard override on analytical application.
6. **Explicit boundary conditions.** AXIOM is not applied universally. It is inappropriate when emotional containment is required, when creative looseness is primary, when intuition should lead, or when cognitive load is high. Restraint is part of the framework. If AXIOM application increases anxiety, rigidity, or cognitive load, it is being misapplied and should be suspended.

2.2 System Definition Stage

Before the core analytical loop is entered, AXIOM requires explicit system definition. This stage establishes the boundary within which all subsequent analysis operates and prevents the most common analytical failure mode: reasoning about a system whose boundary has never been stated.

The system definition stage specifies:

- **System of Interest (SoI):** what is inside the boundary.
- **Environment:** what is outside the boundary but interacts with the system.
- **Interfaces:** what crosses the boundary (inputs, outputs, actors).
- **Measured variable(s):** what is actually tracked or scored in practice.

- **Valued variable(s):** what the system is ultimately intended to produce or protect.
- **Time horizon:** the distance over which “success” is being evaluated.

The explicit separation of *measured* and *valued* variables at the definition stage is the primary structural intervention AXIOM makes relative to unstructured analysis. This separation is the precondition for the Goodhart check in the core loop: if the distinction is not established at the outset, metric substitution failures are invisible by default.

2.3 The Core Analytical Loop

The core loop consists of ten steps applied in sequence after the system definition stage is complete. Each step is mandatory; steps may be brief when the domain is simple, but none may be omitted.

1. **Invariants.** What must always be true for this system to remain the system of interest? Invariants are structural constraints that, if violated, mean the system has changed into a different system. Identifying invariants bounds the space of valid interventions.
2. **Constraints.** What are the rules or limits that govern the system? Constraints may be physical, regulatory, resource-based, or normative. Unlike invariants, constraints can be relaxed by external decision, but doing so changes what interventions are feasible.
3. **Hypothesis.** What do we think is happening? The hypothesis is a provisional causal account of the system’s current behaviour. It is held explicitly and provisionally, not as a conclusion.
4. **Short feedback loop.** How can the hypothesis be tested quickly and cheaply? The short feedback loop is designed to generate early disconfirming or confirming evidence before significant resources are committed.
5. **Falsifier.** What outcome would prove the hypothesis wrong? The falsifier is stated before any test is run. A hypothesis that cannot produce a falsifier is classified as a closed epistemic system and flagged for domain gating.
6. **Control loop specification.** When a feedback loop is claimed, it must be specified in four components:
 - (a) *Signal* – what is measured;
 - (b) *Comparator* – what the signal is compared against (threshold, target, norm);
 - (c) *Actuator* – what changes behaviour in response to the comparator output;
 - (d) *Delay* – the typical lag between signal and actuator response, and known failure modes when the lag increases. A claimed feedback loop that cannot be specified in these terms is not a control loop, it is an assumption.
7. **Goodhart check (metric substitution audit).** In what ways could an agent “win” the measured variable while the valued variable degrades? What second metric or structural change would reveal gaming? The Goodhart check is applied to the *measured* variable identified in the system definition stage. It is not a speculation about bad actors; it is a structural analysis of whether the measurement architecture is robust to optimisation pressure.
8. **Predictions and perturbation tests.** Two to three explicit predictions are stated in the form “if X, then Y within time horizon T”. In addition, one perturbation test is required: what change would move the system into a qualitatively different operating regime?
9. **Abstraction and transfer.** What structural pattern does this domain instantiate? Is the pattern recognisable in other domains? The abstraction step is the primary mechanism by which AXIOM generates cross-domain leverage. Named structural patterns (Section 2.4) are produced by this step.
10. **Impact evaluation.** After application, the analyst evaluates: Did clarity increase? Did agency increase? Did anxiety decrease? Did rigidity increase? If rigidity increases, the application is suspended and the framework is refined.

2.4 Named Structural Patterns

Extended application of the AXIOM loop across diverse domains has produced a set of named structural patterns: recurring failure architectures that appear independently across operationally distinct systems. These patterns are produced by the abstraction step (Step 9) and represent the primary mechanism by which AXIOM generates cross-domain transfer value. Four patterns are formalised here; all emerged from empirical domain testing rather than being specified in advance.

2.4.1 Nested Goodhart Failure via Compounding Proxy Chain

A standard Goodhart failure occurs when a measured variable is decoupled from the valued variable it proxies: optimising the metric degrades the outcome. A *nested* Goodhart failure occurs when the defence of an intervention refers to a second system whose own measured variable is itself Goodhart-compromised – producing a compounding proxy chain in which the valued variable is doubly (or multiply) removed from any measured variable in the chain.

The pattern was identified during AXIOM analysis of ABA therapy for autism, where the clinical defence via educational integration rights chains two independently Goodhart-compromised systems: the treatment optimises behavioural compliance metrics, and the educational system optimises test performance, while the valued variable (genuine intellectual and developmental flourishing) is the terminal node that no metric in the chain reaches.

Generalisation conditions: the pattern appears wherever (a) an intervention is defended by reference to the system it integrates a subject into, and (b) that receiving system is itself Goodhart-compromised. No single intervention fix is sufficient; the full chain must be restructured.

Confirmed transfers: disaster preparedness (compliance metrics → perceived political competence → actual population resilience); music grade examinations (grade → audition inclusion → professional viability); ABA therapy as described above.

2.4.2 The Crystallised Success Pattern

When a domain produces observable successful outcomes, institutions attempt to reverse-engineer those outcomes into a credentialing system. The credential becomes the measured variable; the actual capability that produced the original successful outcomes becomes the valued variable. The two diverge because the credential system was not the mechanism that produced the original capability, that capability was produced by direct, often informal, engagement with the domain.

The pattern was identified during AXIOM analysis of instrumental music grade examination systems. The credential then functions as an economic gate that:

1. provides revenue justification for the training institution;
2. provides legal and reputational cover for selection decisions;
3. systematically underweights the capability it claims to measure; and
4. requires qualitative human override to correct when the gate produces the wrong outcome.

Pattern conditions: the Crystallised Success Pattern appears wherever

1. a domain produces observable successful outcomes,
2. those outcomes are economically valuable enough to justify a training industry,
3. the industry requires a measurable output to sustain its economic justification, and
4. the underlying capability is qualitative and resistant to direct measurement.

Confirmed transfers: cybersecurity certification (CISSP, CEH and equivalents); legal bar examinations; medical board scores versus clinical judgment; sports coaching qualifications.

Perturbation confirmation: internet distribution disrupted the economic gate layer of music education without re-designing the credentialing system, and the valued variable (musical capability and expression) continued to be produced and distributed outside the system. The perturbation confirmed the structural analysis: the constraint was the gate, not the capability.

2.4.3 Constitutive vs. Contingent Measurement Gaps

Standard AXIOM analysis assumes that measurement/value gaps are *contingent*: they arise from human choices (wrong metric, incentive misalignment, methodological limitation) and are therefore fixable in principle by better design. Extended domain testing identified a second class: *constitutive* gaps, where the gap is physically or logically irreducible and no intervention design can close it.

The distinction was identified during AXIOM analysis of quantum measurement under the Copenhagen interpretation, where the full quantum state (the valued variable) is destroyed by the act of measurement. The Heisenberg uncertainty principle is not a technology limitation; it is a statement that the gap cannot be closed even in principle. The correct prescriptive response differs by gap type:

- **Contingent gap:** redesign the loop, fix the metric.
- **Constitutive gap:** accept epistemic limits; redesign decisions around irreducible uncertainty.

Important constraint: the constitutive classification is itself a falsifiable AXIOM claim. The falsifier is: if the quantum measurement gap can be shown to be contingent (i.e. the full quantum state is determinate prior to measurement under a hidden-variable interpretation), the constitutive classification is wrong. Three active candidates in physics – Bohmian mechanics, the Many-Worlds interpretation, and quantum state tomography – each offer distinct resolutions. The gap-type classification is therefore provisional.

Candidate applications to social science domains: felt belonging, subjective experience, and phenomenological states may be constitutive rather than contingent gaps, suggesting that the correct design response in those domains is different decision architecture, not better instrumentation.

2.4.4 Proxy Chain Audit Procedure

The nested Goodhart pattern requires a corresponding procedural check that is not present in the standard AXIOM loop. The proxy chain audit sits between the Goodhart check (Step 7) and the predictions step (Step 8) and is triggered when an intervention is defended by reference to a right, value, or outcome external to the immediate system.

The procedure is:

1. Identify the rights-based or valued-outcome justification being offered for the intervention.
2. Trace the full chain from the intervention’s measured variable to the claimed valued variable, including all intermediate systems the subject is being integrated into.
3. For each link in the chain, apply the standard Goodhart check: can this link be won while the next link degrades?
4. If any link in the chain is Goodhart-compromised, flag the defence as *incomplete* – it does not cleanly reach the valued variable.

The validation study was designed to test whether AXIOM-structured outputs demonstrate higher measurement integrity and analytical coherence than unstructured baseline outputs, when scored against peer-reviewed domain literature by an independent evaluation pass. The design separates the generation phase from the scoring phase to reduce the risk that domain-specific knowledge available at scoring time contaminates the generation.

2.5 Design Overview

The study comprised two sequential phases.

Phase 1 – Blinded generation. For each domain, a seed paragraph describing a real-world operational problem was provided as input. Two artifacts were generated per domain:

- *A Naive Baseline:* a standard intervention analysis produced without structural scaffolding, representative of unguided LLM reasoning on the same problem.
- *An AXIOM Output:* a structured analysis produced by applying the full AXIOM core loop (Section 2.3), including system definition, invariants, constraints, hypothesis, short feedback loop, falsifier, control loop specification, Goodhart check, predictions, perturbation test, and abstraction.

No web searches were conducted during Phase 1. All generation was performed without access to domain-specific literature, to ensure that the outputs reflected the framework’s structural reasoning rather than retrieval of known answers.

Phase 2 – Literature validation. Following the completion of all 15 Phase 1 artifacts, web searches were conducted for all 15 domains simultaneously. Each domain’s AXIOM output was scored against peer-reviewed literature on the five dimensions described in Section 2.8. Scoring occurred after generation was complete for all domains, preventing knowledge of scoring outcomes from influencing subsequent generation.

2.6 Domain Selection

Fifteen domains were selected to provide operational diversity across problem types, institutions, and physical environments. The selection criteria were: (1) the domain has a well-defined operational problem with a clear measured

variable and a non-identical valued variable; (2) peer-reviewed literature exists against which outputs can be scored; and (3) the domains collectively span biological, physical, organisational, and safety-critical systems. No domain was selected because it was expected to produce a strong AXIOM result.

The domains were:

1. Hospital nursing handoff reliability
2. Antibiotic stewardship and resistance management
3. Shift work fatigue management (mining and transport)
4. Wildfire incident command resource allocation
5. Airport ground operations turnaround
6. Water distribution leak management (non-revenue water)
7. Wastewater biological nutrient removal (BNR) process stability
8. Food safety HACCP execution and critical control point monitoring
9. Perioperative scheduling and operating room throughput
10. Medication administration error prevention
11. Preventive maintenance scheduling for heavy equipment fleets
12. Construction falls-from-height safety controls
13. Cold-chain integrity for vaccines and biologics
14. Noise-induced hearing loss prevention
15. Infection prevention in aged care respiratory outbreaks

2.7 Output Specification

Each AXIOM output was required to address all of the following components, corresponding to the core loop steps described in (Section 2.3).

1. **System Definition:** system of interest, environment, interfaces, measured variable(s), valued variable(s), time horizon.
2. **Invariants:** structural constants that must hold for the system to remain the system of interest.
3. **Constraints:** rules, limits, and resource bounds governing the system.
4. **Hypothesis:** provisional causal account of the system’s current failure mode.
5. **Short Feedback Loop:** proposed fast, cheap test of the hypothesis.
6. **Falsifier:** at least one explicit statement of what outcome would demonstrate the hypothesis is wrong.
7. **Control Loop Specification:** signal, comparator, actuator, and delay for any claimed feedback loop.
8. **Goodhart Check:** at least one way the measured variable could be won while the valued variable degrades, plus a proposed second metric or structural mitigation.
9. **Predictions and Perturbation Test:** two to three “if X, then Y within T” predictions, plus one perturbation test.
10. **Abstraction:** the structural pattern instantiated by the domain, and at least one proposed transfer target.

The Naive Baseline was generated from the same seed paragraph and was required only to address the operational problem in a manner representative of unstructured expert analysis. No structural template was applied to the baseline.

2.8 Scoring Framework

2.8.1 Dimensions

An initial scoring scheme with four equal-weight dimensions was revised after a methodological challenge: prediction accuracy (the ability to specify correct quantitative thresholds) was not identified in the AXIOM specification as a primary design goal. The revised framework introduced a fifth dimension (Falsifier Coherence) and applied differential weights consistent with AXIOM’s stated purpose. The final five dimensions are:

- **MR – Mechanism Recall** (weight $\times 3$): Does the output correctly identify the causal mechanisms described in peer-reviewed literature for this domain? This dimension tests whether AXIOM’s structural analysis converges on the mechanistic account supported by empirical evidence.
- **MIG – Measurement Integrity and Goodhart Handling** (weight $\times 3$): Does the output correctly identify the gap between measured and valued variables, and does the Goodhart check identify realistic and literature-supported ways in which the measured variable can be won while the valued variable degrades?
- **FC – Falsifier Coherence** (weight $\times 2$): Are the stated falsifiers logically consistent with the hypothesis? Would the specified outcome actually falsify the causal claim? Are the falsifiers distinguishable from the predictions (i.e. do they identify disconfirming rather than confirming conditions)?
- **EFC – Experiment-Form Convergence** (weight $\times 2$): Does the proposed short feedback loop, control loop specification, or perturbation test converge with the research designs used in the peer-reviewed literature to study this domain? This dimension tests whether AXIOM generates structurally appropriate empirical forms without access to domain-specific methodology literature.
- **PC – Prediction Consistency** (weight $\times 1$): Are the directional predictions consistent with the empirical literature? This dimension receives the lowest weight because quantitative threshold precision is not a primary AXIOM design goal, and because the blinded generation condition systematically disadvantages precise quantitative prediction.

2.8.2 Scoring procedure

Each dimension was scored on a three-point scale per domain (0 = not present or incorrect, 1 = partially correct, 2 = fully correct and literature-supported), prior to application of dimensional weights. The maximum weighted score per domain is 33 points ($(3 + 3 + 2 + 2 + 1) \times 2 \times \dots$; see Table 1 for the full calculation). The maximum total weighted score across all 15 domains is 495.

Table 1: Weighted scoring formula. Each dimension is scored 0–2 per domain; the weight multiplier is applied before summing.

Dim.	Description	Weight	Max per domain	Max across 15 domains
MR	Mechanism Recall	$\times 3$	6	90
MIG	Measurement Integrity & Goodhart	$\times 3$	6	90
FC	Falsifier Coherence	$\times 2$	4	60
EFC	Experiment-Form Convergence	$\times 2$	4	60
PC	Prediction Consistency	$\times 1$	3	45
Total per domain		–	33	495

2.8.3 Baseline comparison

The Naive Baseline was scored on the same dimensions using the same rubric. Differential scoring (AXIOM minus Baseline) is reported by dimension in Section 3.2 to isolate the contribution of the structural framework from base model capability.

2.9 Methodological Constraints on Record

The following constraints are declared explicitly as part of the study record. They are not presented as deficiencies to be minimised but as epistemic facts that a reader requires in order to correctly interpret the results.

MC-01 – Single unrepeated sample. Inference parameters (temperature, sampling strategy) were not visible or controllable during generation. All results represent a single unrepeated sample per domain. No confidence intervals on generation variance are reportable.

MC-02 – Self-evaluation. Scoring in Phase 2 was conducted by the same system that generated the Phase 1 artifacts. This introduces a self-preference bias risk that cannot be fully eliminated under the present design. The risk is partially mitigated by the use of an external literature corpus as the scoring ground truth (the system cannot score an output as correct unless the claim is supported by peer-reviewed evidence), but the framing and selection of scoring evidence remained under the control of the generating system.

MC-03 – Post-hoc weight revision. The differential weighting scheme described in Section 2.8 was introduced after the methodological challenge regarding prediction accuracy, not pre-registered before Phase 1 generation. The revised weights are justified by the stated design goals of AXIOM (measurement integrity and falsifiability over prediction precision), but the exploratory nature of the weighting revision must be acknowledged. The unweighted baseline score (88%, 158/180) is reported alongside the weighted score for transparency.

MC-04 – Literature access asymmetry. The lowest-performing domain (wildfire incident command resource allocation, 67%) was also the domain with the least accessible peer-reviewed academic literature in Phase 2. It is not possible to determine from the present data whether the lower score reflects a genuine AXIOM failure mode or a scoring artefact of limited literature coverage.

MC-05 – Genuine out-of-sample gap identified. One verifiable knowledge gap was detected in Phase 2: the AXIOM output for cold-chain integrity did not address freeze damage as a second cold-chain failure mode. Vaccine vial monitors (VVMs) register heat exposure only, not freeze-thaw cycles, meaning a vaccine consignment can pass VVM inspection while having been rendered ineffective by freezing. This gap was not identified in the Phase 1 output and is not attributable to the scoring asymmetry noted in MC-02.

3 Results

3.1 Domain-Level Results

Table 2 presents the weighted score for each of the 15 domains. The maximum weighted score per domain is 33 points, derived from the dimensional weights specified in Section 2.8: two dimensions at weight $\times 3$ (maximum 6 points each), two at weight $\times 2$ (maximum 4 points each), and one at weight $\times 1$ (maximum 3 points), each scored on a 0–2 scale before weighting.

Table 2: Domain-level weighted scores (maximum 33 per domain).

No.	Domain	Score	%
1	Hospital nursing handoff reliability	32/33	97%
2	Antibiotic stewardship and resistance management	32/33	97%
3	Shift work fatigue management (mining/transport)	33/33	100%
4	Wildfire incident command resource allocation	22/33	67%
5	Airport ground operations turnaround	27/33	82%
6	Water distribution leak management (NRW)	33/33	100%
7	Wastewater biological nutrient removal (BNR) process stability	30/33	91%
8	Food safety HACCP execution and CCP monitoring	32/33	97%
9	Perioperative scheduling and OR throughput	33/33	100%
10	Medication administration error prevention	33/33	100%
11	Preventive maintenance scheduling for heavy equipment fleets	25/33	76%
12	Construction falls-from-height safety controls	32/33	97%
13	Cold-chain integrity for vaccines and biologics	32/33	97%
14	Noise-induced hearing loss prevention	33/33	100%
15	Infection prevention in aged care respiratory outbreaks	32/33	97%
Total (weighted)		461/495	93%
Total (unweighted baseline)		158/180	88%

Both the weighted and unweighted scores are reported in accordance with Methodological Constraint MC-03 (Section 2.9). The 5 percentage point difference between unweighted (88%) and weighted (93%) scores reflects the differential emphasis on measurement integrity and mechanism recall over prediction consistency – dimensions on which AXIOM performs more strongly by design.

3.2 Dimension-Level Results

Table 3 presents aggregate scores for each dimension across all 15 domains, before and after application of dimensional weights.

Measurement Integrity and Goodhart Handling (MIG) was the strongest dimension (98%). Identification of the gap between measured and valued variables, and the multiple ways each metric could be won while the underlying

Table 3: Dimension-level results across 15 domains.

Dim.	Description	Weight	Raw	Max Raw	%
MR	Mechanism Recall	$\times 3$	42/45	45	93%
MIG	Measurement Integrity & Goodhart	$\times 3$	44/45	45	98%
FC	Falsifier Coherence	$\times 2$	43/45	45	96%
EFC	Experiment-Form Convergence	$\times 2$	41/45	45	91%
PC	Prediction Consistency	$\times 1$	33/45	45	73%
Weighted total		–	461/495		93%

system degrades, was confirmed by peer-reviewed literature across all 15 domains. This result is consistent with AXIOM’s primary design goal: the explicit separation of measured and valued variables at the system definition stage forces this analysis as a structural prerequisite rather than leaving it to emerge incidentally from unstructured reasoning.

Falsifier Coherence scored 96% (FC). Literature supports that correct causal structure – tested by falsifiers – is a prerequisite for valid intervention prediction, and that mechanistic models outperform purely statistical models in extrapolative and out-of-sample scenarios. The FC score suggests that AXIOM’s requirement for at least one explicit falsifier per hypothesis is sufficient to produce causal structures that are consistent with those identified in the domain literature.

Experiment-Form Convergence scored 91% (EFC). The proposed short feedback loops, control loop specifications, and perturbation tests generated by AXIOM converged substantially with the research designs used in the peer-reviewed literature for these domains. The four domains with partial EFC losses (Domains 3, 4, 7, and 11) all involved either specialised measurement infrastructure (e.g., polysomnography for fatigue, SCADA systems for BNR process control) or operational variables (e.g., wildfire resource allocation under non-steady conditions) where the proposed experimental forms were directionally correct but insufficiently specific.

Prediction Consistency was the weakest dimension (73%, PC). This result is expected and not interpreted as a primary failure mode. Directional prediction accuracy was high across domains; losses were predominantly on quantitative threshold precision (e.g., specific percentage reductions in error rates, numerical recurrence intervals for equipment failure). As noted in Section 2.8, quantitative threshold precision is not a primary AXIOM design goal under blinded generation conditions, and PC carries the lowest dimensional weight in the revised rubric for this reason.

3.3 Distribution of Performance

Domain scores were distributed as follows:

- **Perfect score (100%, 33/33):** Domains 3, 6, 9, 10, 14 – 5 of 15 domains (33%).
- **Near-perfect (97%, 32/33):** Domains 1, 2, 8, 12, 13, 15 – 6 of 15 domains (40%).
- **High (91%, 30/33):** Domain 7 – 1 of 15 domains (7%).
- **Moderate (76–82%):** Domains 5 and 11 – 2 of 15 domains (13%).
- **Lower (67%, 22/33):** Domain 4 – 1 of 15 domains (7%).

The right-skewed distribution – with 11 of 15 domains (73%) scoring 97% or above – suggests that the primary sources of variance are concentrated in a small subset of domains and dimensions, rather than being distributed uniformly across the experiment. The three lower-performing domains (4, 5, and 11) share a common feature: their measured-to-valued-variable gaps involve strong operational or institutional factors that are poorly represented in the academic literature (wildfire resource allocation), or where quantitative threshold precision is the limiting factor rather than structural mechanism identification (preventive maintenance, airport turnaround schedule padding).

3.4 Key Quantitative Findings

Four findings are highlighted as the primary quantitative results of the validation experiment.

Finding 1: Measurement integrity was confirmed as a structural strength across all 15 domains. The MIG dimension (98%, 44/45 raw) showed no domain where the AXIOM-identified measured/valued gap was inconsistent with the peer-reviewed characterisation of the domain’s core operational failure mode. This is the strongest single-dimension result and provides the most direct empirical support for AXIOM’s central design claim: that forcing the

measured/valued separation at the system definition stage produces more accurate analytical outputs than unstructured reasoning on the same problem.

Finding 2: A cross-domain structural invariant was confirmed. Across Domains 1, 8, 10, 11, 12, and 15, the same structural failure pattern appeared independently: a *completion marker* (form submitted, record closed, observation filed) is decoupled from the work it certifies and is set by the party under production or accountability pressure. The corrective architecture is identical across all six cases: granular step-level sign-off, automated record generation, or third-party witnessing at the point of work completion rather than at the point of administrative closure. This cross-domain invariant was identified by AXIOM’s abstraction step (Step 9) and confirmed by literature in each domain independently. It provides empirical support for AXIOM’s structural transfer claim: that domain-independent patterns can be identified through structured reasoning and subsequently confirmed in domain-specific evidence.

Finding 3: The lowest-performing domain (wildfire resource allocation, 67%) was also the domain with the least accessible peer-reviewed academic literature. The primary losses were on MR (Mechanism Recall) and EFC (Experiment-Form Convergence), consistent with limited literature coverage rather than structural framework failure. The MIG score for Domain 4 remained high (the measured/valued gap identification was correct); the losses were in mechanism-level specificity and experimental form. This finding is consistent with Methodological Constraint MC-04, which notes that it is not possible to determine from the present data whether the lower score reflects a genuine AXIOM failure mode or a scoring artefact of limited literature coverage.

Finding 4: One genuine out-of-sample knowledge gap was identified (Methodological Constraint MC-05). The AXIOM output for Domain 13 (cold-chain integrity) did not address freeze damage as a second cold-chain failure mode. Vaccine vial monitors (VVMs) register heat exposure only; they do not detect freeze-thaw cycles, meaning a vaccine consignment can pass VVM inspection while having been rendered ineffective by freezing [Matthias et al., 2007]. This gap was not present in the Phase 1 output and represents a genuine limitation of the blinded generation approach: AXIOM’s Goodhart check identified the VVM-as-measured-variable gap correctly, but the perturbation test did not surface freeze damage as an alternative failure mode at the actuator level.

4 Discussion

The results presented in Section 3.4 support three broad claims: that AXIOM’s measurement integrity protocol produces analytical outputs that converge with peer-reviewed domain literature at high rates; that the abstraction step produces genuine cross-domain structural patterns rather than superficial analogies; and that the framework’s primary failure modes are predictable from its design constraints rather than arbitrary. This section develops the implications of each claim in turn, then addresses extended domain testing and the framework’s implications for LLM reasoning architecture and organisational systems analysis.

4.1 Cross-Domain Structural Invariant

The most theoretically significant result of the validation experiment was not the overall weighted score but the identification of a repeating structural failure pattern across six operationally unrelated domains: hospital nursing handoff (Domain 1), HACCP execution (Domain 8), medication administration error prevention (Domain 10), preventive maintenance scheduling (Domain 11), construction falls-from-height safety (Domain 12), and infection prevention in aged care (Domain 15).

In each of these domains, independently and without prior knowledge of the others, the following structural failure appears in the peer-reviewed literature: a *completion marker*; a form submission, a record closure, a checklist sign-off, an observation log entry; is decoupled from the work it certifies, and is set by the party operating under production or accountability pressure. The measured variable (the marker) can be won while the valued variable (work quality, safety, patient outcome) degrades. The corrective architecture identified in the literature is identical across all six cases: granular step-level sign-off at the point of work completion, automated record generation, or third-party witnessing that separates the attestation function from the production function.

This pattern constitutes what AXIOM’s Step 9 (Abstraction) is designed to surface: a structural invariant that transcends domain. Its independent confirmation in six peer-reviewed literatures under blinded generation conditions is not a trivial result. It suggests that the structural failure – administrative closure decoupled from task completion – is a genuine attractor in operational system design rather than a domain-specific phenomenon. The corrective architecture’s invariance across domains further suggests that the fix is similarly structural: once the decoupling is identified and the attestation function is separated from the production function, the same design solution applies regardless of domain.

The cross-domain structural invariant supports a broader claim about AXIOM’s abstraction step: that it functions as a *pattern detection mechanism* rather than merely a reflective exercise. The pattern was not specified in advance, was not present in the seed paragraphs provided to the framework, and was not derived from any single domain’s literature. It emerged from the repeated application of the same structural analysis to operationally diverse systems. This is precisely the transfer mechanism that AXIOM’s design anticipates.

4.2 Extended Domain Testing

Following the primary validation experiment, AXIOM was applied in extended domain testing across four additional domains in March 2026. These applications are not part of the primary validation and were not scored against the rubric described in Section 2.8. They are reported here as qualitative extensions that generated candidate additions to the framework’s named pattern library (Section 2.4).

4.2.1 Nested Goodhart Failure and the Proxy Chain Audit

Analysis of ABA therapy for autism yielded the *nested Goodhart failure via compounding proxy chain* pattern. The clinical defence of the intervention via educational rights legislation chains two independently Goodhart-compromised systems: the treatment optimises behavioural compliance metrics, and the educational system optimises test performance, while the valued variable (genuine intellectual and developmental flourishing) is the terminal node that no metric in the chain reaches.

This pattern differs from single-layer Goodhart failure in a theoretically important respect: no single intervention fix is sufficient. Addressing the Goodhart failure at the treatment layer does not resolve the failure at the receiving-system layer. The full chain must be restructured. The proxy chain audit procedure (Section 2.4) was developed as a direct procedural response to this pattern.

The pattern was confirmed across three additional domains: disaster preparedness policy (political accountability metrics → actual population resilience), music grade examination systems (grade → audition inclusion → professional viability), and ABA therapy as described above. These transfers are documented in Section 2.4 and Appendix 6.

4.2.2 Constitutive vs. Contingent Measurement Gaps

Analysis of quantum measurement under the Copenhagen interpretation generated a taxonomic distinction not present in the primary validation: the difference between *contingent* gaps (arising from human design choices, fixable in principle) and *constitutive* gaps (physically or logically irreducible, where the measurement act destroys the valued variable).

All 15 primary validation domains produced contingent gaps. The quantum measurement case is the first candidate constitutive gap encountered in extended testing. The practical implication of the distinction is that the prescriptive response differs by gap type: contingent gaps call for redesign, while constitutive gaps call for decision architecture that accepts irreducible uncertainty as a structural feature rather than a problem to be solved.

This classification is itself a falsifiable AXIOM claim. The constitutive reading of the quantum measurement gap fails if Bohmian mechanics (non-local hidden variables), the Many-Worlds interpretation, or ensemble-level quantum state tomography can be shown to make the gap contingent at the individual system level. The gap-type classification is therefore held provisionally and is flagged as a candidate addition to AXIOM’s system definition stage in a future version.

4.2.3 The Crystallised Success Pattern

Analysis of instrumental music grade examination systems generated the *Crystallised Success Pattern*: the observation that when a domain produces observable successful outcomes, institutions reverse-engineer those outcomes into a credentialling system. The credential becomes the measured variable; the capability that produced the original successful outcomes becomes the valued variable; the two diverge because the credentialling system was not the mechanism that produced the original capability.

A perturbation test confirmed this analysis empirically. Internet distribution disrupted the economic gate layer of music education without redesigning the credentialling system, and the valued variable (musical capability and expression) continued to be produced and distributed outside the system. The perturbation confirmed that the constraint on capability production was the gate, not the capability itself.

The pattern transferred to four additional domains without modification: cybersecurity certification (CISSP, CEH and equivalents), legal bar examinations, medical board scores versus clinical judgment, and sports coaching qualifications.

In each case, the same structure was observed: observable successful outcomes $\rightarrow$ economic justification for training institution $\rightarrow$ credential as measured variable $\rightarrow$ qualitative human override required to correct the credential gap at the point of selection.

4.3 Implications for LLM Reasoning Architectures

The validation results have direct implications for the use of structured reasoning protocols in LLM systems, independent of AXIOM specifically.

Measurement integrity as a typed output contract. The strongest result in the experiment; 98% on MIG across 15 domains; suggests that requiring an explicit separation of measured and valued variables as a structural output requirement (rather than an emergent property of unstructured reasoning) reliably produces outputs that converge with domain-expert characterisations of the core operational failure mode. This has a direct architectural implication: measurement integrity can be enforced as a *typed output contract* in LLM systems, analogous to structured output schemas, without requiring domain-specific fine-tuning. The constraint is structural, not parametric.

Falsification as a governance layer. The 96% score on Falsifier Coherence across domains that include both physical safety systems and organisational processes suggests that the falsification requirement generalises across domain types. This is consistent with the epistemological literature on falsifiability [Popper, 1959], which holds that falsifiability is a property of the statement structure rather than the domain content. An LLM constrained to produce at least one explicit falsifier per hypothesis will systematically produce output that is more auditable and more resistant to unfalsifiable reasoning patterns; the structural signature of epistemic closure identified in Working Thread 4 of the AXIOM project as a prerequisite for value-system abuse.

The Goodhart check as a reasoning audit. The validation results suggest that requiring an explicit Goodhart check; a statement of how the measured variable could be won while the valued variable degrades; does not require domain expertise to produce outputs that are consistent with domain literature. This is a potentially valuable finding for LLM reasoning architecture: it suggests that Goodhart resistance can be built into reasoning structure rather than into domain-specific training, and that a generalist reasoning layer can produce Goodhart-aware outputs across operationally diverse domains.

Limitations of blinded generation for quantitative prediction. The 73% score on Prediction Consistency is the direct cost of blinded generation: a reasoning framework that does not have access to domain-specific numerical data will systematically underperform on quantitative threshold prediction relative to domain-expert baselines. This is a known limitation of pure structural reasoning and does not represent a failure of the framework for its intended purpose. However, it does indicate that AXIOM is not a substitute for domain expertise in tasks where quantitative precision is the primary requirement. The appropriate architectural position is as a *pre-analysis layer* that structures the problem before domain-specific data is applied, rather than as a replacement for empirical domain knowledge.

4.4 Implications for Organisational Systems Analysis

Beyond LLM reasoning architecture, the results have implications for the use of structured frameworks in operational and organisational settings.

The completion-marker invariant as an audit heuristic. The cross-domain structural invariant identified in Section 4.1; administrative closure decoupled from task completion, set by the party under accountability pressure; is directly actionable as an organisational audit heuristic. Any system in which a completion marker is self-reported by the party whose performance is being assessed carries the structural precondition for this failure mode. Auditors, safety regulators, and operations managers can apply this check independently of domain expertise: identify who sets the marker, identify who benefits from the marker being set, and check whether the attestation function is structurally separated from the production function.

The proxy chain audit as a policy evaluation tool. The nested Goodhart failure pattern and the proxy chain audit procedure have direct applications in policy evaluation. Any policy intervention that is defended by reference to a right, valued outcome, or second-order system should be subjected to a proxy chain audit before acceptance. The formal validity of a right-based defence does not guarantee that the defence reaches the valued variable; it guarantees only that the proximate metric is covered by a normative framework. These are different claims, and conflating them is a systematic source of policy failure that the proxy chain audit is designed to surface.

Restraint as a design constraint. AXIOM’s sixth design principle (biological constraint layer) and seventh (explicit boundary conditions) encode the view that a reasoning framework that does not specify its own non-use conditions is incomplete. This has a practical organisational implication: frameworks that are applied universally tend to become

ideology. The explicit boundary conditions; emotional containment required, creative looseness primary, intuition leading, cognitive load high; define the space where AXIOM’s application would increase rigidity rather than clarity. An organisation adopting structured analytical frameworks should specify analogous non-use conditions to prevent the framework from becoming a cognitive constraint that outlasts its utility.

5 Limitations

This section consolidates the methodological constraints declared in Section 2.9 into a unified limitations statement and assesses their implications for the interpretation of the results.

Self-evaluation bias (MC-02). The most significant limitation of the present study is that scoring in Phase 2 was conducted by the same system that generated the Phase 1 artifacts. The literature on LLM-as-judge methodology documents systematic self-preference bias at the individual output level [Zheng et al., 2023]. The present design partially mitigates this by anchoring scoring to an external peer-reviewed literature corpus – the system cannot score an output as correct unless the claim is supported by a source it did not generate – but the framing of search queries, the selection of which sources to prioritise, and the interpretation of partial matches all remained under the control of the generating system. This bias is not resolvable within the present experimental design. Replication with an independent human expert panel scoring blinded outputs against the same rubric is the appropriate follow-on study.

Single unrepeated sample (MC-01). All 15 Phase 1 outputs represent single generation samples under uncontrolled inference parameters. No estimate of generation variance is available. It is not possible to determine whether the results represent typical outputs of the framework applied to these domains or whether they are drawn from the tail of the generation distribution in either direction. Monte Carlo replication across multiple generation samples per domain, with variance reporting by dimension, would substantially strengthen the reliability claims.

Post-hoc weight revision (MC-03). The differential weighting scheme was introduced after a methodological challenge identified prediction accuracy as inconsistent with AXIOM’s stated design goals, not before Phase 1 generation. The revised weights are justified by the framework specification, but the post-hoc nature of the revision means the weighted score of 93% should be treated as exploratory rather than confirmatory. The unweighted baseline score (88%) is reported throughout as a transparency measure. Pre-registration of the weighting scheme before generation is required for the results to carry confirmatory weight in future iterations.

Literature access asymmetry (MC-04). The lowest-performing domain (wildfire incident command resource allocation, 67%) was also the domain with the least accessible peer-reviewed academic literature. The causal direction of the association between literature scarcity and lower scores cannot be determined from the present data. Future studies should pre-screen domains for literature coverage and report it as a covariate.

Genuine out-of-sample gap (MC-05). The cold-chain freeze damage gap (Section 3.4, Finding 4) is a genuine knowledge limitation of the blinded generation approach. It establishes that AXIOM’s Goodhart check and perturbation test are not exhaustive: they correctly identify the primary measured/valued gap but may not surface all alternative failure modes at the actuator level. The appropriate response is not to treat this as a scoring loss only, but as a specification for a directed extension: perturbation tests should explicitly probe *second failure modes* at each control loop layer, not only the primary Goodhart failure.

Scope of the Naive Baseline comparison. The present study reports AXIOM output scores against peer-reviewed literature, not differential scores against the Naive Baseline on a dimension-by-dimension basis. Baseline dimension-level data were collected but are not reported in full here due to space constraints and the self-evaluation limitation noted above. Full differential reporting is deferred to a follow-on study with independent scoring.

Generalisability across model architectures. All outputs were generated by a single model architecture under a single prompting regime. It is not known whether the results generalise to other architectures, parameter scales, or prompting implementations of the AXIOM protocol. The framework is specified as a *typed output contract* (Section 3.4), which in principle should be architecture-agnostic, but empirical cross-architecture replication has not been conducted.

6 Conclusion

This paper introduced AXIOM, a falsification-first reasoning framework designed to improve measurement integrity in structured analysis across operationally diverse domains. The framework enforces explicit separation of measured and valued variables, requires falsifiable hypotheses, specifies control loops, and audits for metric substitution risk at each analytical step.

A blinded generation experiment across 15 domains, followed by post-hoc scoring against peer-reviewed literature using a differential-weight rubric, produced an overall weighted score of 93% (461/495), with the strongest performance on Measurement Integrity and Goodhart Handling (98%) and Falsifier Coherence (96%), and the weakest on Prediction Consistency (73%) – a dimension not central to the framework’s design purpose.

Three substantive findings emerged from the experiment. First, the measurement integrity result – 98% across all 15 domains under blinded generation – provides empirical support for the claim that forcing a structured measured/valued separation at the system definition stage reliably produces outputs that converge with domain-expert characterisations of the core operational failure mode. Second, the cross-domain structural invariant (administrative completion markers decoupled from task completion, confirmed across six independent domains) demonstrates that AXIOM’s abstraction step produces genuine transferable patterns rather than superficial analogies. Third, one genuine out-of-sample gap was identified (cold-chain freeze damage), which provides a concrete specification for a directed extension of the perturbation test procedure.

Extended domain testing produced three candidate additions to the framework’s named pattern library: the nested Goodhart failure via compounding proxy chain, the Crystallised Success Pattern, and the constitutive versus contingent measurement gap distinction. These patterns are held provisionally and are flagged for formalisation in a later version of AXIOM, pending additional domain testing and independent validation.

The framework’s primary architectural contribution is the Goodhart check as a structural requirement rather than an emergent property. Existing structured prompting methods – chain-of-thought, ReAct, Reflexion, Constitutional AI – optimise for task performance or harmlessness but do not enforce measurement integrity as a typed output property. The present results suggest that this requirement can be added as a structural constraint without domain-specific tuning and without degrading mechanism recall or falsifier coherence. The appropriate position for AXIOM in an LLM reasoning architecture is as a pre-analysis layer that structures problem representation before domain-specific data or tools are applied.

Significant methodological limitations constrain the present findings. Self-evaluation bias, single-sample generation, and post-hoc weight revision collectively mean the quantitative results should be treated as exploratory. The appropriate follow-on study requires independent human expert scoring of blinded outputs, pre-registered weighting, and Monte Carlo replication across generation samples. The present paper establishes the experimental protocol and baseline results for that follow-on study.

AXIOM is explicitly provisional. Its design principles include provisionality as a first-order constraint: the framework is a testing scaffold, not a worldview, and it is subject to revision by falsifying evidence. The most important such evidence would be a domain in which systematic application of AXIOM increases analytical rigidity rather than clarity – the failure condition specified in the framework’s own impact evaluation step. No such domain was encountered in the primary validation, but the experimental design was not structured to surface it. Identifying AXIOM’s genuine failure domains is as important as confirming its successes, and it is the next priority for structured empirical investigation.

Appendix A: Domain Validation Literature

The following sources constituted the peer-reviewed corpus used in Phase 2 scoring. For each domain, sources are listed in the order they were consulted. Where a source informed scoring on multiple dimensions, the relevant dimensions are noted in parentheses (MR = Mechanism Recall; MIG = Measurement Integrity and Goodhart Handling; FC = Falsifier Coherence; EFC = Experiment-Form Convergence; PC = Prediction Consistency).

Domain 1: Hospital Nursing Handoff Reliability

- Buffum, M.D., Blegen, M.A., & Lyndon, A. (2015). Handoffs and patient safety: Grasping the story and painting a full picture. *Western Journal of Nursing Research*, 37(11), 1458–1478. *PMC4272331*. (MR, MIG, EFC)
- Toren, O., Lipschuetz, M., Lehmann, A., Regev, G., & Arad, D. (2022). Improving patient safety in general hospitals using structured handoffs: Outcomes from a national project. *Frontiers in Public Health*, 10, 777678. (MR, EFC)
- Smeulders, M., Lucas, C., & Vermeulen, H. (2014). Effectiveness of different nursing handover styles for ensuring continuity of information in hospitalised patients. *Cochrane Database of Systematic Reviews*, 6, CD009979. (MR, FC, PC)
- Agency for Healthcare Research and Quality. (2023). Communication during transitions of care. *PSNet*. (MIG, EFC)

- Kim, J.H., Lee, J.L., & Kim, E.M. (2021). Patient safety culture and handoff evaluation of nurses in small and medium-sized hospitals. *International Journal of Nursing Sciences*, 8(1), 80–86. *PMC7859534*. (MIG, PC)

Domain 2: Antibiotic Stewardship and Resistance Management

- Centers for Disease Control and Prevention. (2026). *Antibiotic use and stewardship in the United States, 2025 update: Progress and opportunities*. CDC. (MR, MIG)
- Apisarnthanarak, A., & colleagues. (2023). Association between antimicrobial stewardship programs and antibiotic use globally: A systematic review and meta-analysis. *JAMA Network Open*, 6(2), e2253806. (MR, FC, PC)
- Chen, W., Zeng, Y., Zheng, J., et al. (2026). Evaluation of antimicrobial resistance governance across 193 countries to inform the 2026 Global Action Plan update. *Nature Medicine*. <https://doi.org/10.1038/s41591-026-04257-1> (MIG, EFC)
- Access to Medicine Foundation. (2026). *2026 Antimicrobial resistance benchmark report*. (MIG, PC)
- StatPearls. (2023). Antimicrobial stewardship. NCBI Bookshelf, *NBK572068*. (MR)

Domain 3: Shift Work Fatigue Management (Mining/Transport)

- Bauerle, T., & Poplin, G. (2018). Mineworker fatigue: A review of what we know and future decisions. *Minerals*, 8(5), 193. *PMC5983045*. (MR, MIG, FC)
- Samaha, E., Lal, S., Samaha, N., & Wyndham, J. (2007). Psychological, lifestyle and coping contributors to chronic fatigue in shift-worker nurses. *Journal of Advanced Nursing*, 59(3), 221–232. (MR, EFC)
- National Institute for Occupational Safety and Health. (2014). Application of fatigue management systems: Small mines and low technology solutions. *PMC4539295*. (EFC, PC)
- Akerstedt, T. (2003). Shift work and disturbed sleep/wakefulness. *Occupational Medicine*, 53(2), 89–94. (MR, FC)

Domain 4: Wildfire Incident Command Resource Allocation

- Cova, T.J., Theobald, D.M., Norman, J.B., & Siebeneck, L.K. (2013). Wildfire evacuee shadow evacuation: Anticipating evacuee behaviour to improve evacuation response. *Environmental Hazards*, 12(2), 94–108. (MR)
- Katuwal, H., Calkin, D., & Hand, M. (2016). Production and efficiency of large wildfire suppression effort: A stochastic frontier analysis. *Forest Policy and Economics*, 67, 1–10. (MIG, PC)
- Note: Domain 4 returned the lowest score (67%) partly due to limited accessible peer-reviewed academic literature on operational ICS resource allocation decisions under fire; see Methodological Constraint MC-04.

Domain 5: Airport Ground Operations Turnaround

- Malighetti, P., Morlotti, C., Redondi, R., & Paleari, S. (2022). The turnaround tactic and on-time performance: Implications for airlines' efficiency. *Research in Transportation Business & Management*, 45, 100874. (MR, MIG)
- Rodríguez-Sanz, Á., et al. (2020). A novel approach for turnaround time allocation. *Proceedings of the 32nd Congress of the International Council of the Aeronautical Sciences*. (MR, EFC)
- ACI Europe. (2025). *Airport performance position paper*. (MIG, PC)
- Airport Cooperative Research Program. (2016). *Common performance metrics for airport infrastructure and operational planning* (ACRP Report 190). Transportation Research Board. (MIG, EFC)

Domain 6: Water Distribution Leak Management (NRW)

- Lambert, A., & Hirner, W. (2000). *Losses from water supply systems: Standard terminology and recommended performance measures*. IWA Publishing. (MR, MIG)
- Farley, M., & Trow, S. (2003). *Losses in water distribution networks*. IWA Publishing. (MR, FC)
- Thornton, J., Sturm, R., & Kunkel, G. (2008). *Water loss control* (2nd ed.). McGraw-Hill. (MR, EFC)
- Wu, Z.Y., & Sage, P. (2006). Pressure dependent demand optimization for leakage detection and pipe burst detection. *Proceedings of the 8th Annual Water Distribution Systems Analysis Symposium*. (EFC, PC)

Domain 7: Wastewater Biological Nutrient Removal (BNR)

- Metcalf & Eddy / Tchobanoglous, G., Stensel, H.D., Tsuchihashi, R., & Burton, F. (2014). *Wastewater engineering: Treatment and resource recovery* (5th ed.). McGraw-Hill. (MR, MIG)
- Henze, M., Gujer, W., Mino, T., & van Loosdrecht, M. (2000). *Activated sludge models ASM1, ASM2, ASM2d and ASM3*. IWA Publishing. (MR, FC)
- Flores-Alsina, X., et al. (2012). Benchmarking biological nutrient removal in wastewater treatment plants: Influence of mathematical model assumptions. *Water Science and Technology*, 65(8), 1496–1505. (EFC, PC)

Domain 8: Food Safety HACCP Execution and CCP Monitoring

- Mortimore, S., & Wallace, C. (2013). *HACCP: A practical approach* (3rd ed.). Springer. (MR, MIG)
- Ropkins, K., & Beck, A.J. (2000). Evaluation of worldwide approaches to the use of HACCP to control food safety. *Trends in Food Science & Technology*, 11(1), 10–21. (MIG, FC)
- Panisello, P.J., & Quantick, P.C. (2001). Technical barriers to hazard analysis critical control point (HACCP). *Food Control*, 12(3), 165–173. (MIG, EFC)
- World Health Organization. (2003). *Hazard analysis and critical control point (HACCP) system and guidelines for its application*. WHO. (MR)

Domain 9: Perioperative Scheduling and OR Throughput

- Vladu, A., & Ghitea, T.C., et al. (2024). Enhancing operating room efficiency: The impact of computational algorithms on surgical scheduling and team dynamics. *PMC11476208*. (MR, EFC)
- Cardoen, B., Demeulemeester, E., & Beliën, J. (2010). Operating room planning and scheduling: A literature review. *European Journal of Operational Research*, 201(3), 921–932. (MR, MIG)
- El Omri, A., et al. (2025). Exploring the landscape of operating room scheduling: A bibliometric analysis of recent advancements and future prospects. *PMC12358003*. (MIG, PC)
- Marjamaa, R., Vakkuri, A., & Kirvela, O. (2008). Operating room management: Why, how and by whom? *Acta Anaesthesiologica Scandinavica*, 52(5), 596–600. (MIG, EFC)

Domain 10: Medication Administration Error Prevention

- Al Mutair, A., et al. (2021). The effective strategies to avoid medication errors and improving reporting systems. *Medicines*, 8(9), 46. *PMC8468915*. (MR, EFC)
- Agency for Healthcare Research and Quality. (2023). Medication administration errors. *PSNet*. (MR, MIG)
- StatPearls. (2023). Medication dispensing errors and prevention. NCBI Bookshelf, *NBK519065*. (MR, FC)
- Bates, D.W., et al. (1999). The costs of adverse drug events in hospitalized patients. *JAMA*, 277(4), 307–311. (MIG, PC)
- Poon, E.G., et al. (2010). Effect of bar-code technology on the safety of medication administration. *New England Journal of Medicine*, 362(18), 1698–1707. (EFC, PC)

Domain 11: Preventive Maintenance Scheduling for Heavy Equipment

- Jardine, A.K.S., Lin, D., & Banjevic, D. (2006). A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7), 1483–1510. (MR, MIG)
- Wang, W. (2012). An overview of the recent advances in delay-time-based maintenance modelling. *Reliability Engineering & System Safety*, 106, 165–178. (MR, FC)
- Mobley, R.K. (2002). *An introduction to predictive maintenance* (2nd ed.). Butterworth-Heinemann. (MR, EFC)
- Note: Domain 11 scored 76% (moderate). Quantitative threshold prediction was the primary gap; directional mechanism recall was confirmed across sources.

Domain 12: Construction Falls-from-Height Safety Controls

- Nadhim, E.A., Hon, C., Xia, B., Stewart, I., & Fang, D. (2016). Falls from height in the construction industry: A critical review of the scientific literature. *International Journal of Environmental Research and Public Health*, 13(7), 638. (MR, MIG)
- Huang, X., & Hinze, J. (2003). Analysis of construction worker fall accidents. *Journal of Construction Engineering and Management*, 129(3), 262–271. (MR, FC)
- Safe Work Australia. (2020). *Construction work: Code of practice*. Safe Work Australia. (EFC)
- Choudhry, R.M., & Fang, D. (2008). Why operatives engage in unsafe work behavior: Investigating factors on construction sites. *Safety Science*, 46(4), 566–584. (MIG, EFC)

Domain 13: Cold-Chain Integrity for Vaccines and Biologics

- World Health Organization. (2015). *Immunization in practice: A practical guide for health staff*. WHO. (MR, MIG)
- Matthias, D.M., Robertson, J., Garrison, M.M., Newland, S., & Nelson, C. (2007). Freezing temperatures in the vaccine cold chain: A systematic literature review. *Vaccine*, 25(20), 3980–3986. (MR, FC) [Note: This source identified the MC-05 gap – freeze damage not detected by VVMs.]
- PATH. (2014). *Vaccine cold chain equipment optimisation platform*. PATH. (EFC)
- Kartoglu, U., & Milstien, J. (2014). Tools and approaches to ensure quality of vaccines throughout the cold chain. *Expert Review of Vaccines*, 13(7), 843–854. (MIG, EFC)

Domain 14: Noise-Induced Hearing Loss Prevention

- Nelson, D.I., Nelson, R.Y., Concha-Barrientos, M., & Fingerhut, M. (2005). The global burden of occupational noise-induced hearing loss. *American Journal of Industrial Medicine*, 48(6), 446–458. (MR, MIG)
- Occupational Safety and Health Administration. (2002). *OSHA technical manual: Noise*. OSHA. (MR, EFC)
- Berger, E.H. (2003). *The noise manual* (5th ed.). American Industrial Hygiene Association. (MR, FC)
- Tak, S., Davis, R.R., & Calvert, G.M. (2009). Exposure to hazardous workplace noise and use of hearing protection devices among US workers – NHANES, 1999–2004. *American Journal of Industrial Medicine*, 52(5), 358–371. (MIG, PC)

Domain 15: Infection Prevention in Aged Care Respiratory Outbreaks

- Lansbury, L.E., Brown, C.S., & Nguyen-Van-Tam, J.S. (2017). Influenza in long-term care facilities. *Influenza and Other Respiratory Viruses*, 11(5), 356–366. (MR, MIG)
- Australian Commission on Safety and Quality in Health Care. (2020). *ACSQHC infection prevention and control*. ACSQHC. (MR, EFC)
- Mody, L., et al. (2015). Multidrug-resistant organisms in long-term care: An updated systematic review. *Annals of Long-Term Care*, 23(1). (MR, FC)
- Eilers, R., et al. (2015). Impact of influenza on elderly people in long-term care facilities: A retrospective database analysis. *Vaccine*, 33(11), 1339–1343. (MIG, PC)

References

- BackOnTruck et al. AXIOM: Benchmarking LLM-as-a-judge for code via rule-based perturbation and multisource quality calibration. arXiv:2512.20159, 2024.
- Y. Bai et al. Constitutional AI: Harmlessness from AI feedback. arXiv:2212.08073, 2022.
- S. M. Barnett and S. J. Ceci. When and where do we apply what we learn? a taxonomy for far transfer. *Psychological Bulletin*, 128(4):612–637, 2002.
- S. Beer. *Brain of the Firm*. Allen Lane, 1972.
- D. T. Campbell. Assessing the impact of planned social change. *Evaluation and Program Planning*, 2(1):67–90, 1979.
- P. Checkland. *Systems Thinking, Systems Practice*. Wiley, 1999.
- D. Gentner and K. J. Holyoak. Reasoning and learning by analogy. *American Psychologist*, 52(1):32–34, 1997.

- C. A. E. Goodhart. Problems of monetary management: The U.K. experience. *Papers in Monetary Economics*, 1, 1975.
- D. Hendrycks et al. Measuring massive multitask language understanding. arXiv:2009.03300, 2021.
- T. Kojima et al. Large language models are zero-shot reasoners. arXiv:2205.11916, 2022.
- V. Krakovna et al. Specification gaming: The flip side of AI ingenuity. In *DeepMind Blog*, 2020.
- M. Levy et al. Signaling and meaning in organizational analytics: Coping with Goodhart’s law in an era of digitization and datafication. *Frontiers in Communication*, 2023. PMC10376445.
- S. Lin et al. TruthfulQA: Measuring how models mimic human falsehoods. arXiv:2109.07958, 2022.
- D. Manheim and S. Garrabrant. Categorizing variants of Goodhart’s law. arXiv:1803.04585, 2019.
- D. M. Matthias, J. Robertson, M. M. Garrison, S. Newland, and C. Nelson. Freezing temperatures in the vaccine cold chain: A systematic literature review. *Vaccine*, 25(20):3980–3986, 2007.
- D. H. Meadows. *Thinking in Systems: A Primer*. Chelsea Green Publishing, 2008.
- B. Miranda et al. Putnam-AXIOM: A functional & static benchmark for measuring higher level mathematical reasoning in LLMs. arXiv:2508.08292, 2025.
- OpenAI. Structured outputs. <https://platform.openai.com/docs/guides/structured-outputs>, 2024.
- L. Podlesnik et al. The influence of causal loop diagrams on systems thinking and information utilization in complex problem-solving. *ResearchGate*, 2025.
- K. R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.
- P. Schmid et al. Reward hacking as equilibrium under finite evaluation. arXiv:2603.28063, 2026.
- P. M. Senge. *The Fifth Discipline: The Art and Practice of the Learning Organization*. Doubleday, 1990.
- N. Shinn et al. Reflexion: Language agents with verbal reinforcement learning. arXiv:2303.11366, 2023.
- J. D. Sterman. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw-Hill, 2000.
- M. Strathern. ‘improving ratings’: Audit in the british university system. In *European Review*, volume 5, pages 305–321. Cambridge University Press, 1997.
- L. von Bertalanffy. *General System Theory*. George Braziller, 1968.
- H. von Foerster. Observing systems. In *Observing Systems*. Intersystems Publications, 1981.
- X. Wang et al. Self-consistency improves chain of thought reasoning in language models. arXiv:2203.11171, 2023.
- J. Wei et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- N. Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press, 1948.
- S. Yao et al. ReAct: Synergizing reasoning and acting in language models. arXiv:2210.03629, 2023a.
- S. Yao et al. Tree of thoughts: Deliberate problem solving with large language models. arXiv:2305.10601, 2023b.
- R. Zellers et al. HellaSwag: Can a machine really finish your sentence? arXiv:1905.07830, 2019.
- L. Zheng et al. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023.